

Academic Elective Bidding & Allocation System (Problem Statement #05)

Deliverable 1: Complete Requirements Specification Table (5 Functional & 2 Non-Functional Requirements)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (Justification)	Comments
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately (sum == 100) and prerequisites validated. Fail: Total credits != 100 or unmet prerequisites bypassed.	Core bidding functionality ensuring fair distribution of student preferences according to academic eligibility.	Base requirement from problem statement specification.
FR-002	Functional	The system shall allow the Academic Registrar to configure elective course offerings, section capacities (min/max seats), and designated timetable slot constraints.	High	Pass: Course records saved with valid capacities and slot times. Fail: Duplicate slot assignments without multi-section flag or invalid seat caps.	Establishes boundary conditions and timetable constraints required by the automated allocation solver.	Validates non-negative seat counts and department slot matrix.
FR-003	Functional	The system shall execute the multi-criteria optimization solver to allocate courses based on bid weightings, CGPA tie-breakers, and room/slot collision avoidance rules.	High	Pass: All eligible bids evaluated; seats filled up to max limit with 0 timetable clashes. Fail: Student assigned to overlapping lecture slots or oversubscribed course.	Core algorithmic automation eliminating manual clash resolution and ensuring objective allocation.	Solver handles hard constraints (no clashes) & soft constraints (preferences).
FR-004	Functional	The system shall allow students to view real-time provisional seat demand percentages and adjust/reallocate their submitted bidding points prior to the deadline window closure.	Medium	Pass: Bid updates reflected instantly in student profile and draft pool before lock time. Fail: Modifications permitted after deadline timestamp.	Provides transparency on course popularity, allowing students to strategize point allocation dynamically.	Access locked immediately upon bidding window expiry timestamp.
FR-005	Functional	The system shall publish final allocation results, generate individualized clash-free timetables for students, and export master enrollment rosters for the Academic Registrar.	Medium	Pass: Downloadable PDF/ICS schedule produced with 0 slot clashes; Registrar roster matches allocated student IDs. Fail: Incomplete roster export or unassigned enrolled courses.	Closes the registration lifecycle with actionable student schedules and official registrar records.	Supports PDF schedule download and Excel/CSV roster export.
NFR-001	Non-Functional (Performance)	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm execution time < 30 seconds for 5,000 concurrent student bids under simulated peak load. Fail: Solver execution exceeds 30 seconds or server runs out of memory.	Prevents server bottlenecks and ensures rapid turnaround between bidding closure and timetable publishing.	Benchmark under synthetic 5,000 student dataset with 40 elective offerings.
NFR-002	Non-Functional (Security & Integrity)	The system shall enforce Role-Based Access Control (RBAC) and maintain an immutable audit trail of all bid submissions, modifications, and administrative overrides.	High	Pass: Unauthorized role access blocked (HTTP 403); 100% of bid modification events logged with timestamp, user ID, and prior state. Fail: Unauthenticated bid injection or unlogged state change.	Guarantees academic integrity, prevents tampering, and ensures auditability during student grievance redressals.	Audit log entries secured with write-once access policies.